Empirical Work: Econometrics 1
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April 20, 2026

1 Theory Questions

1.1 OLS Bias

The potential model with which we are working is as follows:

$$Y = \beta_0 + \beta_1 X + h(W) + U, \tag{1}$$

where Y is the baby weight, X is a dummy whether the mother smoked during pregnancy and W is a vector of controls.
Running an OLS model of Y on X and a constant gives us:

$$Y = \gamma_0 + \gamma_1 X + \epsilon \tag{2}$$

By OLS we have $\mathbb{E}[\epsilon] = \mathbb{E}[\epsilon X] = 0$. The coefficient of X will be:

$$\gamma_1 = \frac{\text{Cov}(Y, X)}{\text{Var}(X)} = \beta_1 + \frac{\text{Cov}(X, U)}{\text{Var}(X)} \tag{3}$$

Since X is endogenous to the potential model, this occurs because the decision to smoke is not random — mothers who smoke may differ from non-smokers in unobserved ways (e.g., socioeconomic background, general health behavior, diet) that independently affect birth weight. These unobservables are captured in U, creating a correlation between X and U, i.e., $\text{Cov}(X, U) \neq 0$. Moreover, we can assume that mothers who smoke tend to have, on average, worse unobserved health behavior and lower socioeconomic status - both which independently reduce birth weight. This means U tends to be lower for smokers, so $\text{Cov}(X, U) < 0$. It gives a biased coefficient, probably lower than β_1, which does not allow us to recover the causal effect of smoking during pregnancy on baby weight.[1]
Even controlling for observable characteristics W, unobserved confounders are likely to remain, so the CIA would be a strong and implausible assumption in this context.

1.2 IV with binary smoking

Now we have a binary instrument, Z, that says whether the mother is in a support group. It is safer to assume that we have heterogeneous effects in this case, since different mothers likely respond differently to smoking. Thus, for the instrument to be considered a good instrument, we need relevance, independency, exclusion, and monotonicity.

For relevance, we need to be in the support group, on average, to change the behavior of mothers about smoking, i.e., $\mathbb{E}[X|Z = 1] - \mathbb{E}[X|Z = 0] \neq 0$. It is a plausible assumption, since support groups tend to instruct the mother that this behavior could be bad for the child. In addition, the support group could even help with smoking detoxification, helping the mother to not smoke consistently during pregnancy.

[1] The term $h(W)$ does not appear explicitly because, when W is omitted from the regression, its effect is absorbed into the residual $\epsilon = h(W) + U - \mathbb{E}[h(W)|X]$. The bias term $\frac{\text{Cov}(X,\epsilon)}{\text{Var}(X)}$ therefore captures both the endogeneity from U and any correlation between X and the omitted W.

The independency and exclusion conditions are the most problematic in this case. Independency says that participation in a support group should be randomly assigned between smokers and non-smokers, i.e., $(Y(1), Y(0), X(1), X(0)) \perp Z$. The exclusion condition says that the instrument should only affect birth weight by changing smoking behavior ($Y(x, z) = Y(x)$).

For the first part, support groups are more targeted to vulnerable moms. It is safe to assume that women with more vulnerable background (e.g. worse socioeconomic status) are more likely to smoke during pregnancy. It would violate the independency condition. To tackle this problem we could check for baseline balance between $Z = 0$ and $Z = 1$ groups on pre-determined variable, such as age, education and socioeconomic status.

Furthermore, once in the support group, the women are instructed about baby health in general, not only about smoking, for example encouraging mother to have prenatal care. It could affect baby weight even if the mother continues to smoke, invalidating the exclusion condition. To address this we could look for evidence that the support group affects outcomes other than through smoking. Since we have prenatal visits data, we could test whether Z affects prenatal visits conditional on smoking status. If it does, that's evidence of a direct channel violating exclusion.

Finally, for the monotonicity condition, we would need that no woman starts to smoke because she was assigned to the support group, i.e., $\Pr(X(1) \leq X(0)) = 1$. It is a reasonable assumption, since it is hard to imagine a support group encouraging smoking.

1.3 Multivalued treatment

With this in mind we can think in a potential outcomes model as follows:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + h(W) + U, \tag{4}$$

where X_1, X_2 and X_3 are dummies for whether the mother smoked during 1, 2 or 3 trimesters of pregnancy, respectively. However, the IV considers only the binary dummy $X = 1$ (smoked at all), collapsing all positive smoking levels into $X = 1$. The IV estimator is then:

$$\hat{\beta}_{IV} = \frac{\mathbb{E}[Y|Z = 1] - \mathbb{E}[Y|Z = 0]}{\mathbb{E}[X|Z = 1] - \mathbb{E}[X|Z = 0]} \tag{5}$$

This creates two related problems. First, the instrument shifts mothers across multiple margins simultaneously, some compliers make large change ($3 \to 0$) while others make small changes ($1 \to 0$), yet all appear identically as X going from 1 to 0. The IV estimand therefore becomes a weighted average of going, $3 \to 0$, $2 \to 0$ and $1 \to 0$, where the weights reflect the proportion of compliers at each margin. This weighted average is difficult to interpret as a single causal parameter.

Second, and maybe more importantly, the binary X is blind to partial reductions in smoking. Mothers who reduce from $3 \to 2$, $3 \to 1$ and $2 \to 1$ trimesters due to the support group are still counted as $X = 1$ both before and after the instrument, so their treatment change is entirely invisible to the IV estimator. Those compliers are ignored, even though their partial behavioral response may carry meaningful causal information.

Thus, even setting aside the threats to the IV assumptions discussed in 1.2, using a binary X in a world where smoking is multivalued means the IV estimator identifies a complicated, hard-to-interpret weighted average of heterogeneous effects, and misses an entire class of compliers altogether.

1.4 Multiple dummies + IV

Now we consider the same model in equation (4). It still is not a good idea to estimate the model with OLS, since the X-variables are still endogenous. The problems cited in part (1.1), mothers who smoked more trimesters likely differ systematically from those who smoke less in unobserved ways that also affect birth weight.

Furthermore, we could not estimate this model with only one IV. In this case we have 3 endogenous variables (X_1, X_2, X_3), but only 1 instrument (Z), so the order condition for identification requires at least as many instruments as endogenous variables. The system is underidentified. That is, with one instrument, we can only trace out one dimension of variation in the first stage. We need independent variation to separately identify β_1, β_2 and β_3, but a single binary Z cannot provide that. One way to rescue identification is to impose restrictions on the β's. Specifically, assuming constant marginal effects ($\beta_1 = \beta_2 = \beta_3$).

1.5 Constant marginal effects

Now we impose constant marginal effects, that is, $\beta_1 = \beta_2 = \beta_3 = \beta$. This means that the effect of smoking one additional trimester is the same regardless of the current level of smoking — going from $0 \to 1$ has the same effect as going from $1 \to 2$ or $2 \to 3$. Under this assumption the model collapses to:

$$Y = \beta_0 + \beta(X_1 + X_2 + X_3) + h(W) + U \tag{6}$$

Defining $S = (X_1 + X_2 + X_3)$ as the number of trimesters smoked, the model becomes:

$$Y = \beta_0 + \beta S + h(W) + U \tag{7}$$

Now we have one endogenous variable (S) and one instrument (Z), so the order condition is satisfied and IV identifies β. The IV estimator is:

$$\hat{\beta}_{IV}^{const} = \frac{\mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0]}{\mathbb{E}[S|Z=1] - \mathbb{E}[S|Z=0]} \tag{8}$$

Despite being convenient, the assumption is likely too strong. There are at least two reasons to doubt it. First, the first trimester may be the most harmful, since it coincides with the critical period of fetal development — suggesting the effect of $0 \to 1$ could be larger than $1 \to 2$ or $2 \to 3$. Second, effects could instead be cumulative and accelerating, with each additional trimester compounding the damage from the previous ones. In either case, imposing a single β masks potentially important heterogeneity in the dose-response relationship: if the true marginal effects differ across smoking levels, the IV estimator identifies a weighted average of β_1, β_2 and β_3, with weights depending on the distribution of compliers across margins, rather than any single interpretable causal parameter.

1.6 Relaxing constant effects

Now we relax the assumption that marginal effects are constant, that is, $\beta_1 \neq \beta_2 \neq \beta_3$. The model is now as in equation 4. However, the instrument only affects mothers who would otherwise smoke exactly 1 trimester, moving them from $1 \to 0$. This means there is only one group of compliers. Table 1 shows all possible groups and, whether ruled out, the condition responsible.

Group	Group	S(1)	S(0)	Possible	By Condition
Never-Takers	NT	0	0	Yes	-
-	-	0	1	No	Monotonicity
-	-	0	2	No	Monotonicity
-	-	0	3	No	Monotonicity
Compliers	C	1	0	Yes	-
Always-Takers 1st Level	AT_1	1	1	Yes	-
-	-	1	2	No	Monotonicity
-	-	1	3	No	Monotonicity
-	-	2	0	No	Assumption 1.6
-	-	2	1	No	Assumption 1.6
Always-Takers 2nd Level	AT_2	2	2	Yes	-
-	-	2	3	No	Monotonicity
-	-	3	0	No	Assumption 1.6
-	-	3	1	No	Assumption 1.6
-	-	3	2	No	Assumption 1.6
Always-Takers 3rd Level	AT_3	3	3	Yes	-

Table 1: Possible Groups in 1.6 Setting

Thus, the active groups are $G = \{NT, C, AT_1, AT_2, AT_3\}$. Applying Law of Iterated Expectation:

$$\mathbb{E}[Y|Z=i] = \mathbb{E}[\mathbb{E}[Y|Z=i, G]] = \sum_{g \in G} \Pr(G=g)\mathbb{E}[Y|Z=i, G=g] \tag{9}$$

Taking the difference $\mathbb{E}[Y|Z=1]-\mathbb{E}[Y|Z=0]$, all terms corresponding to NT, AT_1, AT_2, AT_3 cancel, since these groups are unaffected by the instrument. Using the exclusion restriction and the complier restriction from this question:

$$\mathbb{E}[Y|Z=1]-\mathbb{E}[Y|Z=0]=\Pr(C)\mathbb{E}[Y(0)-Y(1)|C]=\Pr(C)*\beta_1 \tag{10}$$

Similarly, since each complier reduces S by exactly 1 and all other groups are unaffected:

$$\mathbb{E}[S|Z=1]-\mathbb{E}[S|Z=0]=\Pr(C)*1 \tag{11}$$

Therefore, the IV estimator identifies:

$$\hat{\beta}_{IV}\xrightarrow{p}\beta_1, \tag{12}$$

the marginal effect of going from 1 trimester to 0. The linear model $Y=\alpha+\beta S+\epsilon$ is misspecified in this context, it imposes a single slope β across all smoking margins, but the true effects are heterogeneous. Since the instrument only moves compliers along the $1\to 0$ margin, IV recovers β_1 only and tells us nothing about β_2 or β_3.

1.7 Caetano & Escanciano method

Now we relax the assumption that the instrument only affects mothers who would otherwise smoke 1 trimester. In this case, the instrument Z can move mothers across any smoking margin, and marginal effects are heterogeneous ($\beta_1\neq\beta_2\neq\beta_3$). As shown in 1.4, the model:

$$Y=\beta_0+\beta_1X_1+\beta_2X_2+\beta_3X_3+h(W)+U,$$

is underidentified with a single binary instrument under classical IV. We now show how the method of Caetano and Escanciano (2021) recovers β_1, β_2 and β_3 by exploiting heterogeneity in the first stage across the controls W.

The instrument Z may affect mothers differently depending on their observable characteristics W. For example, among less educated mothers the support group may induce many to go from $3\to 0$, generating a large first stage for X_3. Among more educated mothers, it may primarily induce $1\to 0$ transitions, generating a large first stage for X_1. This heterogeneity in how Z shifts smoking behavior across values of W is the source of additional identifying variation.

Under the conditional independency assumption $\mathrm{Cov}(U,Z|W)=0$, we can condition on W and take the difference between $Z=1$ and $Z=0$:

$$\mathbb{E}[Y|W,Z=1]-\mathbb{E}[Y|W,Z=0]=\sum\beta_j(\mathbb{E}[X_j|W,Z=1]-\mathbb{E}[X_j|W,Z=0])=\sum\beta_j\Delta_j \tag{13}$$

where $\Delta_j(W)\equiv\mathbb{E}[X_j|W,Z=1]-\mathbb{E}[X_j|W,Z=0]$ is the first stage for X_j conditional on W, for $j=1,2,3$. Thus, we have:

$$\mathbb{E}[Y|W,Z=1]-\mathbb{E}[Y|W,Z=0]=\beta_1\Delta_1(W)+\beta_2\Delta_2(W)+\beta_3\Delta_3(W) \tag{14}$$

Denoting the left-hand side as $\Delta_Y(W)$, we obtain one equation per value of W:

$$\Delta_Y(W)=\beta_1\Delta_1(W)+\beta_2\Delta_2(W)+\beta_3\Delta_3(W) \tag{15}$$

This is a system with 3 unknowns $(\beta_1,\beta_2,\beta_3)$ and as many equations as values of W. As long as W takes sufficiently many values and the first stages $\Delta_1(W)$, $\Delta_2(W)$, $\Delta_3(W)$ are linearly independent as functions of W, the system is overidentified and we can recover all three marginal effects. This linear independence condition is what Caetano and Escanciano (2021) call covariance completeness.

The identification strategy translates directly into a TSLS estimator. Consider first stages of the form:

$$X_j=\alpha_{0j}+\alpha_{1j}Z+\boldsymbol{\alpha}_{2j}'W+\boldsymbol{\alpha}_{3j}'(Z\cdot W)+U_j,\qquad j=1,2,3 \tag{16}$$

where $W=(W_1,W_2,\ldots,W_K)'$ is a vector of K controls and the interaction term $\boldsymbol{\alpha}_{3j}'(Z\cdot W)$ captures how the effect of Z on X_j varies across different characteristics. The instruments are $\{Z, Z\cdot W_1, Z\cdot$

$W_2, \ldots, Z \cdot W_K\}$, giving $K+1$ instruments in total for 3 endogenous variables. The order condition requires $K+1 \geq 3$, that is, at least 2 controls in W.

The covariance completeness condition requires:

$$\text{rank} \begin{pmatrix} \alpha_{11} & \alpha_{31,1} & \cdots & \alpha_{31,K} \\ \alpha_{12} & \alpha_{32,1} & \cdots & \alpha_{32,K} \\ \alpha_{13} & \alpha_{33,1} & \cdots & \alpha_{33,K} \end{pmatrix} = 3 \tag{17}$$

where $\alpha_{3j,k}$ denotes the coefficient on $Z \cdot W_k$ in the first stage for X_j. That is, the first stages must respond to Z in linearly independent ways across the controls. If all three smoking dummies responded proportionally to Z for every value of W, their individual effects could not be separated.

In our empirical setting this is plausible, since we have multiple controls available — education, age group, and prenatal visits — so W is multidimensional and the rank condition can reasonably hold. Under this condition, the TSLS estimator treating X_1, X_2, X_3 as endogenous, W as controls, and $\{Z, Z \cdot W\}$ as instruments consistently estimates β_1, β_2 and β_3.

1.8 Assumptions and tests

The identification strategy from 1.7 rests on three assumptions. We discuss each in turn, along with potential threats and empirical tests.

Assumption 1: Conditional Exogeneity. The first assumption is:

$$\text{Cov}(U, Z|W) = 0 \tag{18}$$

This requires that, conditional on the controls W, the instrument Z is uncorrelated with the unobservable determinants of birth weight U. Since the support group was randomly assigned, this assumption holds unconditionally, and therefore also conditionally on W, making it the most credible of the three. However, if assignment was not purely random and more vulnerable mothers were systematically selected into the support group, then even after conditioning on observed W, unobserved differences between the $Z = 0$ and $Z = 1$ groups may remain, violating exogeneity.

To tackle this concern, we could perform baseline balance tests, comparing pre-determined characteristics such as age, education and prenatal care across the $Z = 0$ and $Z = 1$ groups. Under true random assignment, these should be statistically indistinguishable. All variables needed for this test are available in the dataset.

Assumption 2: Covariance Completeness. The second assumption is the rank condition:

$$\text{rank} \begin{pmatrix} \alpha_{11} & \alpha_{31,1} & \cdots & \alpha_{31,K} \\ \alpha_{12} & \alpha_{32,1} & \cdots & \alpha_{32,K} \\ \alpha_{13} & \alpha_{33,1} & \cdots & \alpha_{33,K} \end{pmatrix} = 3$$

This requires that the first stages for X_1, X_2, X_3 respond to Z in linearly independent ways across values of W. Intuitively, the support group must shift different smoking margins differently depending on mother characteristics, it cannot affect all smoking levels proportionally regardless of W.

The main threat is that the support group affects all mothers similarly regardless of their characteristics, generating proportional first stages and collapsing the rank condition. In this case the method reduces back to the underidentified classical IV problem from 1.4.

Importantly, this assumption is directly testable in the data. We can estimate the first stage regressions for X_1, X_2, X_3 including $Z \times W$ interactions and test whether the interaction coefficients $\alpha_{3j,k}$ are jointly significant and linearly independent across $j = 1, 2, 3$. Moreover, since in our setting we have multiple controls, education, age group and prenatal visits, we have more instruments than endogenous variables, which allows for standard overidentification tests to assess the internal consistency of the moment conditions.

Assumption 3: Structural Separability. The third assumption is that W enters the structural equation additively and separably from the endogenous variables:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + h(W) + U \tag{19}$$

This rules out interactions between W and the X_j's in the structural equation, meaning the marginal effects β_1, β_2, β_3 are constant across values of W. If the effect of smoking genuinely varies with mother characteristics, for example, if smoking during the first trimester is more harmful for older mothers, then separability fails and the β_j's are not well-defined constants.

To assess this, we can test separability by augmenting the structural equation with interactions $W_k \times X_j$ and testing whether their coefficients are jointly zero, using the overidentifying instruments as a basis for the test. This is discussed explicitly as a robustness check in Caetano and Escanciano (2021). The data required for this test, smoking levels interacted with education and age group, are already available in the dataset.

2 Empirical report

2.1 Introduction

Understanding the health consequences of maternal smoking during pregnancy is an important question in both economics and public health. A large medical literature documents that smoking during pregnancy is associated with lower birth weight, which is in turn linked to infant mortality, childhood development, and long-term health outcomes. Estimating the causal effect of smoking on birth weight, however, is challenging: mothers who smoke are likely to differ from non-smokers in unobserved ways (such as socioeconomic background and general health behavior) that independently affect birth weight, making simple comparisons between smokers and non-smokers uninformative about causality.

A further complication arises from the multivalued nature of the treatment. Smoking during pregnancy is not a binary decision (mothers may smoke during one, two, or all three trimesters) and the marginal effect of each additional trimester of smoking need not be the same. Standard instrumental variable (IV) methods are unable to separately identify these marginal effects when only a single instrument is available, typically forcing researchers to either use a binary treatment variable or impose the strong assumption that marginal effects are constant across all smoking levels.

This empirical section of this work addresses both challenges using a dataset of 440,856 birth records from a Brazilian city, combined with the method of Caetano and Escanciano (2021). We exploit the random assignment of pregnant mothers to a pregnancy support group as an instrument for smoking behavior, and apply the Caetano and Escanciano (2021) identification strategy to recover separate marginal effects for each trimester of smoking using a single binary instrument. Our main finding is that marginal effects are strongly heterogeneous: smoking during the first trimester reduces birth weight by approximately 0.57 kg, while smoking during the third trimester reduces it by only 0.10 kg. The null hypothesis of constant marginal effects is decisively rejected by the data.

The remainder of the paper is structured as follows. Section 2.2 describes the data and presents descriptive statistics. Section 2.3 outlines the empirical strategy. Section 2.4 presents the main results and robustness checks. Section 2.5 concludes.

2.2 Data Description and Descriptive Statistics

The dataset contains 440,856 birth records from a random Brazilian city. Table 2 describes all variables used in the analysis. The outcome variable Y is birth weight in kilograms. The treatment variable X is the number of trimesters the mother smoked during pregnancy, taking values in $\{0, 1, 2, 3\}$. The instrument Z is a binary variable indicating whether the mother was randomly offered participation in a pregnancy support group. The controls W include the mother's highest level of education (Middle School, High School, College, or Masters), her age group (15-20, 21-25, 26-30, or 31-40 years), and the number of prenatal visits during pregnancy, which takes only two values (8 or 16) and is recoded as a binary dummy equal to one for mothers with 16 visits.

Table 3 presents descriptive statistics for all variables. The average birth weight is 2.55 kg, with a standard deviation of 0.60 kg, indicating substantial heterogeneity across births. The average number of trimesters smoked is 1.41, suggesting that the average mother in this sample smoked during more than one trimester of pregnancy. The support group instrument is split almost exactly 50/50 across the sample (mean = 0.499), consistent with the random assignment design. The sample is predominantly young, 81% of mothers are between 15 and 25 years old, and relatively low-educated, with 48% holding only a middle school degree.

Table 2: Variable Description

Role	**Variable**	**Description**	**Units / Values**
Outcome (Y)	Baby weight	Birth weight of the baby	Kilograms (continuous)
Treatment (X)	Smoked trimesters	Number of trimesters the mother smoked during pregnancy	0, 1, 2, or 3 (discrete)
Instrument (Z)	Support group	Whether the mother was randomly offered participation in a pregnancy support group	0 = not offered, 1 = offered (binary)
Control (W)	Education	Highest level of education attained by the mother	Middle School / High School / College / Masters
Control (W)	Age group	Age group of the mother at time of birth	15–20 / 21–25 / 26–30 / 31–40 (years)
Control (W)	Prenatal visits	Number of prenatal medical visits during pregnancy	8 or 16 (binary)

Note: Data from a random city in Brazil. N = 440,856 births.

Table 3: Descriptive Statistics (N = 440,856)

Variable	Mean	SD	Min	Max
Baby weight (kg)	2.548	0.601	1.500	5.200
Smoked trimesters	1.406	1.041	0.000	3.000
Support group (Z)	0.499	0.500	0.000	1.000
Prenatal visits (high)	0.434	0.496	0.000	1.000
Age: 15–20	0.391	0.488	0.000	1.000
Age: 21–25	0.420	0.494	0.000	1.000
Age: 26–30	0.083	0.276	0.000	1.000
Age: 31–40	0.105	0.307	0.000	1.000
Education: Middle School	0.483	0.500	0.000	1.000
Education: High School	0.259	0.438	0.000	1.000
Education: College	0.148	0.355	0.000	1.000
Education: Masters	0.110	0.313	0.000	1.000

Table 4 presents the balance test, comparing pre-determined characteristics across support group assignment ($Z = 0$ vs $Z = 1$). This test provides evidence on the plausibility of the independence assumption discussed in Section 1.2 and the first assumption from Section 1.8. For all age group and education variables, differences between the two groups are statistically insignificant at conventional levels, with p-values ranging from 0.106 to 0.970. This strongly supports the random assignment of the instrument and the validity of the independence assumption.

Table 4: Balance Test: Pre-determined Characteristics by Support Group Assignment. P-values from two-sided t-tests.

Variable	No support (Z=0)	Support (Z=1)	Difference	P-value
Prenatal visits (high)	0.405	0.463	0.058	<0.001
Age: 15–20	0.392	0.390	-0.002	0.121
Age: 21–25	0.419	0.422	0.002	0.106
Age: 26–30	0.083	0.084	0.000	0.600
Age: 31–40	0.106	0.105	-0.001	0.544
Education: Middle School	0.482	0.483	0.000	0.801
Education: High School	0.259	0.260	0.000	0.742
Education: College	0.148	0.147	-0.001	0.427
Education: Masters	0.110	0.110	0.000	0.970

The only variable that is significantly imbalanced across groups is prenatal visits. Mothers assigned to the support group have a 5.8 percentage point higher rate of high prenatal visits (46.3% vs 40.5%), a difference that is highly statistically significant ($p < 0.001$). This finding has important implications for the empirical strategy.

There are two possible interpretations of this imbalance. The first is that prenatal visits are exogenous, determined independently of support group assignment, in which case including prenatal visits as a control is harmless and simply absorbs variation in birth weight unrelated to smoking. Under this interpretation, controlling for prenatal visits improves precision without introducing bias.

The second and more concerning interpretation is that prenatal visits are a *mediator*, that is, the support group causes mothers to attend more prenatal visits, which in turn improves birth weight independently of any change in smoking behavior. Under this interpretation, prenatal visits is a bad control in the sense of conditioning on it blocks part of the causal channel from Z to Y, violating the exclusion restriction and biasing the IV estimates upward in absolute value.

The evidence from the balance test points toward the second interpretation. The statistically significant imbalance in prenatal visits across $Z = 0$ and $Z = 1$ groups suggests that support group assignment directly influences prenatal care behavior, consistent with the support group instructing mothers about the importance of prenatal visits beyond just smoking cessation. As discussed in Section 1.2, this represents a direct channel from Z to Y that does not operate through smoking, threatening the exclusion restriction.

In light of this concern, we present our main results both with and without prenatal visits as a control variable. If the two specifications yield similar estimates, the concern is empirically minor. If they differ substantially, this serves as evidence that the exclusion restriction is violated through the prenatal care channel, and our IV estimates should be interpreted with caution.

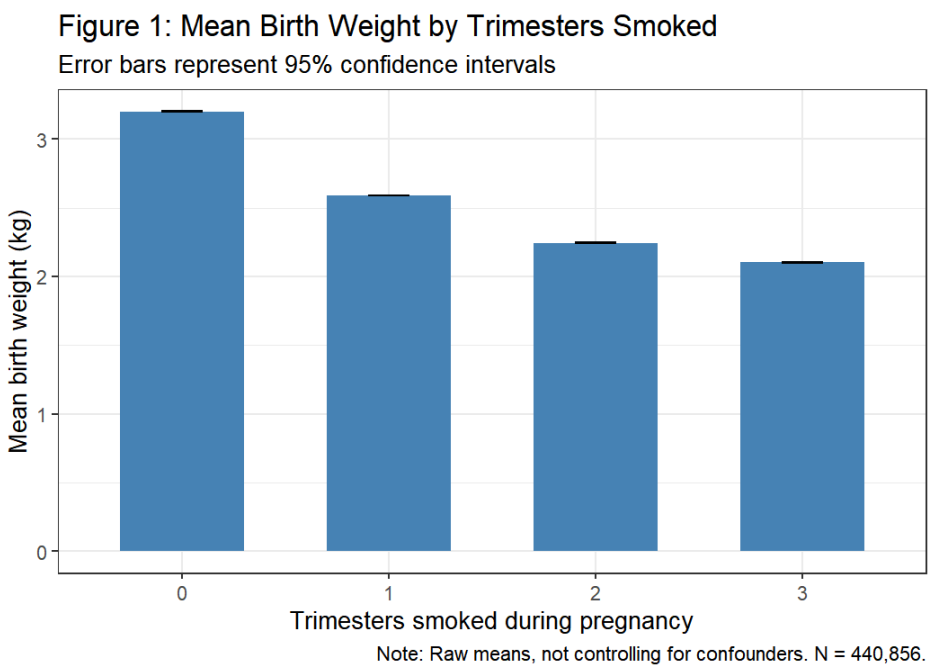

Figure 1: Mean birth weight by number of trimesters smoked during pregnancy. Error bars represent 95% confidence intervals. Raw unconditional means, not controlling for confounders. N = 440,856.

Figure 1 provides a first look at the relationship between smoking and birth weight. It plots mean birth weight by number of trimesters smoked, with 95% confidence intervals. Three patterns stand out. First, there is a clear negative relationship between smoking and birth weight across all levels — mothers who smoked more trimesters consistently had lighter babies. Second, and most importantly for our analysis, the relationship is strongly *non-linear*: the drop in mean birth weight from 0 to 1 trimester smoked (approximately 0.61 kg) is nearly twice as large as the drop from 1 to 2 trimesters (0.35 kg), which is in turn more than twice as large as the drop from 2 to 3 trimesters (0.14 kg). Third, the confidence intervals are extremely tight given the large sample size, so this non-linearity is not a

sampling artifact.

These raw patterns already suggest that the marginal effect of smoking one additional trimester is not constant across smoking levels, the first trimester appears substantially more harmful than subsequent ones. This motivates our use of the method of Caetano and Escanciano (2021) to recover separate marginal effects β_1, β_2, and β_3, rather than imposing a single constant effect as in standard IV. We note, however, that Figure 1 reflects raw unconditional means and does not control for confounders, the patterns observed here may partly reflect selection bias, as discussed in Section 1.1. The causal estimates are presented in Section 2.3.

2.3 Empirical Strategy

Our goal is to estimate the causal effect of smoking during pregnancy on birth weight. As discussed in Section 1.1, OLS is inconsistent in this setting because the decision to smoke is not random, mothers who smoke are likely to differ systematically from non-smokers in unobserved ways, such as socioeconomic background and general health behavior, that independently affect birth weight. This generates a negative correlation between the treatment and the error term, causing OLS to overestimate the harm of smoking.

To address this, we use the random assignment to the pregnancy support group as an instrument for smoking behavior. As argued in Section 1.2, the instrument satisfies relevance (the support group significantly reduces smoking) and the independence assumption is supported by the balance test in Table 4, which shows no statistically significant differences in pre-determined characteristics across $Z = 0$ and $Z = 1$ groups. The monotonicity assumption is also plausible, since it is difficult to imagine the support group causing mothers to smoke more.

The main threat to validity is the exclusion restriction. As documented in Table 4 and discussed in Section 2.2, the support group also increases prenatal visits, suggesting a direct channel from Z to Y that does not operate through smoking. To assess the severity of this concern, all specifications are presented both with and without prenatal visits as a control variable in Section 2.4.

Identification Strategy

A key challenge in this setting is that smoking is a multivalued treatment (mothers can smoke during 0, 1, 2, or 3 trimesters) while the instrument is a single binary variable. As shown in Section 1.4, the classical IV order condition fails: we have three endogenous variables (X_1, X_2, X_3) but only one instrument, making the system underidentified under standard IV. One way around this is to impose constant marginal effects ($\beta_1 = \beta_2 = \beta_3$), collapsing the treatment to a single variable S (trimesters smoked). However, as argued in Section 1.5 and suggested by Figure 1, this assumption is likely too strong and masks important heterogeneity in the dose-response relationship.

We therefore use the method of Caetano and Escanciano (2021), which exploits heterogeneity in the first stage across the controls W to separately identify the marginal effects β_1, β_2, and β_3 without imposing constant effects. The key insight is that conditioning on W and taking the difference between $Z = 1$ and $Z = 0$ yields one equation per value of W:

$$\Delta_Y(W) = \beta_1\Delta_1(W) + \beta_2\Delta_2(W) + \beta_3\Delta_3(W), \tag{20}$$

where $\Delta_Y(W) \equiv \mathbb{E}[Y|W, Z = 1] - \mathbb{E}[Y|W, Z = 0]$ is the conditional reduced form and $\Delta_j(W) \equiv \mathbb{E}[X_j|W, Z = 1] - \mathbb{E}[X_j|W, Z = 0]$ is the conditional first stage for X_j. If the first stages $\Delta_1(W)$, $\Delta_2(W)$, $\Delta_3(W)$ are linearly independent as functions of W (the covariance completeness condition) this system can be solved to recover all three marginal effects.

In practice, this is implemented as a TSLS estimator with first stages of the form:

$$X_j = \alpha_{0j} + \alpha_{1j}Z + \boldsymbol{\alpha}'_{2j}W + \boldsymbol{\alpha}'_{3j}(Z \cdot W) + U_j, \quad j = 1, 2, 3, \tag{21}$$

where the instruments are Z and its interactions with all controls W, that is, $\{Z, Z \times \text{education}, Z \times \text{age group}, Z \times \text{prenatal visits}\}$. The second stage then regresses birth weight on the fitted values $\hat{X}_1$, $\hat{X}_2$, $\hat{X}_3$ and the controls W.

First Stage and Instrument Relevance

Table 5 presents the first stage and reduced form estimates. The support group assignment has a large and highly significant effect on smoking behavior: being offered the support group reduces the average number of trimesters smoked by 1.18, from a mean of 2.0 trimesters in the control group to 0.8 in the treatment group. The first stage F-statistic exceeds 200,000, far above any conventional threshold for instrument relevance, ruling out weak instrument concerns entirely.

Table 5: First Stage and Reduced Form Estimates. Dependent variables: trimesters smoked (first stage) and birth weight in kg (reduced form). Robust standard errors (HC1) in parentheses. Baseline: Middle School, age 15–20, 8 prenatal visits.

Variable	First Stage	Reduced Form
Support group (Z)	-1.185 (0.003)	0.433 (0.002)
Education: High School	-0.017 (0.003)	0.077 (0.002)
Education: College	-0.043 (0.004)	0.153 (0.002)
Education: Masters	-0.064 (0.005)	0.217 (0.003)
Age: 21–25	-0.017 (0.003)	-0.007 (0.002)
Age: 26–30	-0.046 (0.005)	-0.000 (0.003)
Age: 31–40	-0.207 (0.004)	0.091 (0.003)
Prenatal visits (high)	-0.001 (0.003)	0.292 (0.002)
N	440,856	440,856
R-squared	0.328	0.216

The reduced form estimate confirms that the support group has a large positive effect on birth weight: being offered the support group increases birth weight by 0.433 kg on average, conditional on controls. This intent-to-treat (ITT) effect reflects the total impact of the support group on birth weight through all channels, both the reduction in smoking and the possible increase in prenatal visits. Taking the ratio of the reduced form to the first stage gives the Wald estimate of $\frac{-0.433}{1.185} \approx -0.365$ kg per trimester, which is the simple IV estimate under the assumption of constant marginal effects.

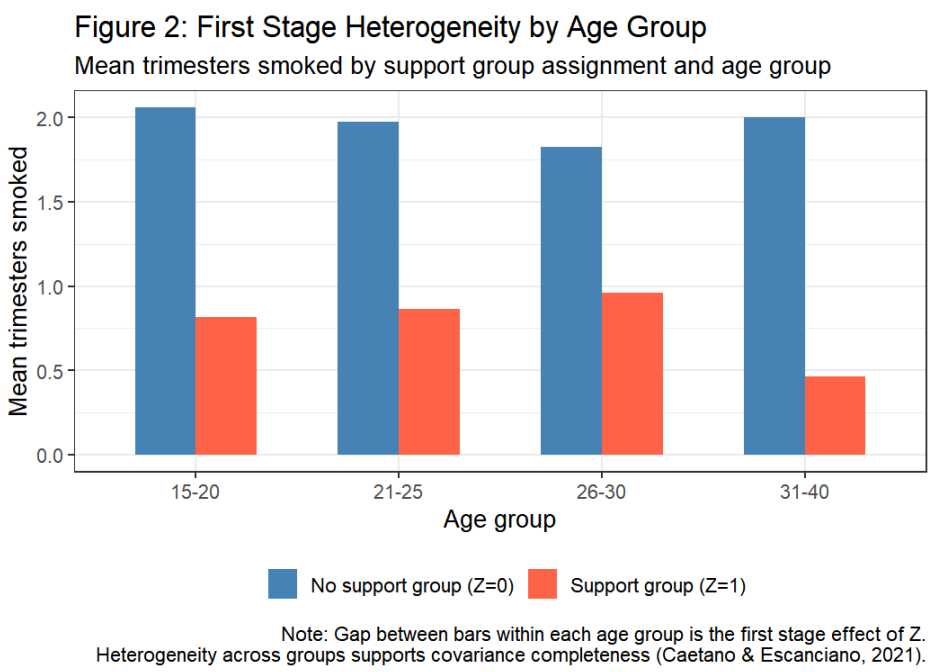

Figure 2: First stage heterogeneity by age group. Mean trimesters smoked by support group assignment and age group. The gap between bars within each age group represents the first stage effect of Z. Heterogeneity across age groups supports the covariance completeness condition of Caetano and Escanciano (2021).

Figure 2 documents the heterogeneity in the first stage across age groups, which is the source of identification in the CE-TSLS approach. The reduction in smoking induced by the support group varies substantially across age groups: it is largest for mothers aged 31–40 (1.54 trimesters) and smallest for mothers aged 26–30 (0.86 trimesters). This variation generates the linearly independent first stages required for covariance completeness. Similar heterogeneity is present across education levels, though less pronounced. Together, these patterns provide empirical support for the rank condition in equation (17).

2.4 Results

Table 6 presents the OLS benchmark estimates. Column (1) regresses birth weight on a binary smoking dummy, yielding an estimate of -0.849 kg for mothers who smoked at any point during pregnancy. Columns (2) and (3) use the number of trimesters smoked as a continuous treatment, imposing constant marginal effects, and yield estimates of approximately -0.364 kg per trimester with and without controls. Column (4) includes the three smoking dummies X_1, X_2, X_3 separately and already reveals substantial heterogeneity in the raw OLS estimates: the effect of starting to smoke ($0 \to 1$ trimester) is -0.600 kg, nearly twice the effect of the second trimester (-0.339 kg) and more than four times the effect of the third (-0.142 kg). However, as discussed in Section 1.1, all OLS estimates are biased due to the endogeneity of smoking: mothers who smoke more are likely to differ from lighter smokers in unobserved ways that independently reduce birth weight, so these estimates should be interpreted as correlations rather than causal effects.

Table 6: OLS Benchmarks: Effect of Smoking on Birth Weight. Dependent variable: birth weight in kilograms. (1) Binary smoking dummy, no controls. (2) Smoking level, no controls. (3) Smoking level, with controls. (4) Three smoking dummies X1, X2, X3, with controls. All OLS estimates are biased due to endogeneity of smoking. Robust standard errors (HC1) in parentheses. Baseline: Middle School, age 15–20, 8 prenatal visits.

Variable	(1)	(2)	(3)	(4)
Smoked (binary)	-0.849 (0.002)			
Smoked trimesters (level)		-0.370 (0.001)	-0.364 (0.001)	
X1: smoked $\geq$ 1 trimester				-0.600 (0.002)
X2: smoked $\geq$ 2 trimesters				-0.339 (0.002)
X3: smoked $\geq$ 3 trimesters				-0.142 (0.002)
Education: High School			0.070 (0.002)	0.066 (0.002)
Education: College			0.138 (0.002)	0.127 (0.002)
Education: Masters			0.194 (0.002)	0.178 (0.002)
Age: 21–25			-0.014 (0.001)	-0.001 (0.001)
Age: 26–30			-0.017 (0.002)	0.003 (0.002)
Age: 31–40			0.015 (0.002)	0.000 (0.002)
Prenatal visits (high)			0.292 (0.001)	0.292 (0.001)
Controls	No	No	Yes	Yes
N	440,856	440,856	440,856	440,856
R-squared	0.358	0.410	0.481	0.517

Table 7 presents the main results, comparing OLS, Simple IV, and CE-TSLS estimates side by side. The Simple IV column instruments the level of smoking S with the support group assignment Z, imposing constant marginal effects as in Section 1.5, and yields an estimate of -0.366 kg per trimester. Notably, this is nearly identical to the OLS estimate in Column (3) of Table 6, suggesting that either selection bias is modest in this dataset, or that the two estimators are picking up similar weighted averages of heterogeneous effects that happen to coincide.

The CE-TSLS estimates, which implement the method of Caetano and Escanciano (2021) as described in Section 2.3, tell a substantially different story. The marginal effect of smoking during the first trimester is $\hat{\beta}_1 = -0.569$ kg ($SE = 0.038$), while the second trimester effect is $\hat{\beta}_2 = -0.383$ kg ($SE = 0.043$), and the third trimester effect is $\hat{\beta}_3 = -0.104$ kg ($SE = 0.045$). These estimates are strikingly heterogeneous: the first trimester is approximately five times more harmful than the third. This pattern is consistent with the biological hypothesis that the first trimester coincides with a critical

Table 7: Main Results: OLS, Simple IV, and CE-TSLS Estimates. Dependent variable: birth weight in kilograms. OLS uses three smoking dummies (X1, X2, X3). Simple IV instruments the level of smoking with Z, imposing constant marginal effects (Section 1.5). CE-TSLS implements Caetano and Escanciano (2021): instruments are Z and $Z \times W$ interactions, recovering separate marginal effects β_1, β_2, β_3. Robust standard errors (HC1) in parentheses. Baseline: Middle School, age 15–20, 8 prenatal visits.

Variable	OLS	Simple IV	CE-TSLS
β_1: smoked $\geq$ 1 trimester	-0.600 (0.002)		-0.569 (0.038)
β_2: smoked $\geq$ 2 trimesters	-0.339 (0.002)		-0.382 (0.043)
β_3: smoked $\geq$ 3 trimesters	-0.142 (0.002)		-0.104 (0.045)
Smoked trimesters (level)		-0.365 (0.001)	
Education: High School	0.066 (0.002)	0.070 (0.002)	0.066 (0.002)
Education: College	0.127 (0.002)	0.138 (0.002)	0.127 (0.003)
Education: Masters	0.178 (0.002)	0.194 (0.003)	0.179 (0.003)
Age: 21–25	-0.001 (0.001)	-0.014 (0.002)	-0.001 (0.002)
Age: 26–30	0.003 (0.002)	-0.017 (0.003)	0.004 (0.004)
Age: 31–40	0.000 (0.002)	0.015 (0.003)	-0.002 (0.003)
Prenatal visits (high)	0.292 (0.001)	0.292 (0.002)	0.292 (0.002)
Method	OLS	2SLS	CE-TSLS
Controls	Yes	Yes	Yes
N	440,856	440,856	440,856

period of fetal development, during which smoking may have a disproportionately large negative effect on birth weight.

The CE-TSLS standard errors are substantially larger than their OLS counterparts, approximately 20 times larger for $\hat{\beta}_1$, which is expected, since CE-TSLS identifies from a narrower source of variation: the heterogeneity in the first stage across values of W, rather than all variation in smoking. Despite the larger standard errors, all three marginal effects remain statistically significant at conventional levels.

Table 8 presents formal tests for the null hypothesis of constant marginal effects. A t-test of $H_0 : \beta_1 = \beta_2$ yields a t-statistic of -2.42 ($p = 0.016$), rejecting equality at the 5% level. A test of $H_0 : \beta_2 = \beta_3$ yields a t-statistic of -3.30 ($p = 0.001$), rejecting at the 1% level. The joint Wald test of $H_0 : \beta_1 = \beta_2 = \beta_3$ yields an F-statistic of 52.49 ($p < 0.001$), strongly rejecting the null of constant marginal effects. These results confirm that the simple IV estimate of -0.366 kg per trimester masks important heterogeneity that is both statistically and economically significant.

Table 8: Tests for Constant Marginal Effects. All tests based on CE-TSLS estimates from Table 7. Tests 1 and 2 are t-tests on pairwise differences of marginal effects, using the joint variance-covariance matrix of the second-stage estimator. Test 3 is a joint Wald F-test of $\beta_1 = \beta_2 = \beta_3$. All three tests reject the null of constant marginal effects at the 5% significance level.

Test	Statistic	Distribution	P-value	Conclusion
$H_0 : \beta_1 = \beta_2$	-2.42	t	0.016	Reject H_0
$H_0 : \beta_2 = \beta_3$	-3.30	t	<0.001	Reject H_0
$H_0 : \beta_1 = \beta_2 = \beta_3$	52.49	$F(2, N-k)$	<0.001	Reject H_0

We also test the structural separability assumption of Caetano and Escanciano (2021), which requires that the marginal effects β_1, β_2, β_3 are constant across values of W. Table 9 shows that augmenting the structural equation with interactions between age group and $\hat{X}_1$ reveals statistically significant coefficients ($p < 0.01$ for all age groups), suggesting that the effect of starting to smoke may vary with the mother's age. This represents a potential limitation of our analysis: if separability fails, the CE-TSLS estimates should be interpreted as population-weighted averages of heterogeneous effects across W, rather than single structural parameters.

Table 9: Separability Test: Interactions between Age Group and $\hat{X}_1$. The structural equation is augmented with interactions between age group and the fitted value of $\hat{X}_1$. Under the separability assumption of Caetano and Escanciano (2021), these coefficients should be jointly zero. Statistically significant interactions suggest the effect of starting to smoke may vary with the mother's age group. Robust standard errors (HC1) in parentheses.

Interaction term	Estimate	Std. Error	t-stat	P-value
Age group 21–25 × $\hat{X}_1$	0.53	0.06	9.05	<0.001
Age group 26–30 × $\hat{X}_1$	-0.22	0.07	-3.15	0.002
Age group 31–40 × $\hat{X}_1$	0.29	0.10	2.80	0.005

Robustness

Table 10 presents the robustness check addressing the bad control concern raised in Section 2.2. Columns (1) and (3) exclude prenatal visits from the set of controls, while Columns (2) and (4) include it. For the Simple IV, excluding prenatal visits changes the estimate from -0.366 to -0.380 kg per trimester, a difference of only 0.014 kg. For the CE-TSLS, the marginal effects are remarkably stable across specifications: $\hat{\beta}_1$ moves from -0.569 to -0.565 kg, $\hat{\beta}_2$ from -0.383 to -0.414 kg, and $\hat{\beta}_3$ from -0.104 to -0.108 kg. These differences are small relative to the standard errors, indicating that the bad control concern has minimal quantitative impact on our conclusions. The finding of strongly heterogeneous marginal effects, with the first trimester substantially more harmful than subsequent ones, holds regardless of how prenatal visits are treated.

Table 10: Robustness Check: With and Without Prenatal Visits as Control. Dependent variable: birth weight in kilograms. Columns (1) and (3) exclude prenatal visits from controls. Columns (2) and (4) include prenatal visits as control. The balance test (Table 4) shows the support group significantly increases prenatal visits, suggesting it may be a mediator rather than an exogenous covariate. The stability of estimates across columns confirms that heterogeneous marginal effects hold regardless of how prenatal visits are treated. Robust standard errors (HC1) in parentheses. Baseline: Middle School, age 15–20.

Variable	Simple IV (1)	Simple IV (2)	CE-TSLS (3)	CE-TSLS (4)
β_1: smoked $\geq$ 1 trimester			-0.565 (0.040)	-0.569 (0.038)
β_2: smoked $\geq$ 2 trimesters			-0.414 (0.044)	-0.382 (0.043)
β_3: smoked $\geq$ 3 trimesters			-0.108 (0.047)	-0.104 (0.045)
Smoked trimesters (level)	-0.380 (0.001)	-0.365 (0.001)		
Education: High School	0.070 (0.002)	0.070 (0.002)	0.066 (0.002)	0.066 (0.002)
Education: College	0.135 (0.003)	0.138 (0.002)	0.125 (0.003)	0.127 (0.003)
Education: Masters	0.193 (0.003)	0.194 (0.003)	0.178 (0.003)	0.179 (0.003)
Age: 21–25	-0.014 (0.002)	-0.014 (0.002)	-0.002 (0.002)	-0.001 (0.002)
Age: 26–30	-0.019 (0.003)	-0.017 (0.003)	0.002 (0.004)	0.004 (0.004)
Age: 31–40	0.012 (0.003)	0.015 (0.003)	-0.004 (0.004)	-0.002 (0.003)
Prenatal visits (high)		0.292 (0.002)		0.292 (0.002)
Prenatal control	No	Yes	No	Yes
N	440,856	440,856	440,856	440,856

2.5 Conclusion

This paper estimates the causal effect of maternal smoking during pregnancy on infant birth weight, using random assignment to a pregnancy support group as an instrument for smoking behavior. Our main contribution is to allow for heterogeneous marginal effects across smoking levels, rather than imposing the constant marginal effects assumption that underlies standard IV approaches.

Using the method of Caetano and Escanciano (2021), which exploits heterogeneity in the first stage across observable controls to separately identify multiple marginal effects with a single binary instrument, we find strong evidence that the dose-response relationship between smoking and birth

weight is highly non-linear. The marginal effect of smoking during the first trimester ($\hat{\beta}_1 = -0.569$ kg) is approximately five times larger in magnitude than the effect of the third trimester ($\hat{\beta}_3 = -0.104$ kg). The null hypothesis of constant marginal effects is rejected by a joint Wald test with an F-statistic of 52.49 ($p < 0.001$). These findings are robust to the inclusion or exclusion of prenatal visits as a control variable, with marginal effect estimates remaining stable across specifications.

These results have important policy implications. Interventions targeting mothers who do not yet smoke, preventing the transition from $S = 0$ to $S = 1$, generate substantially larger health returns per unit of effort than interventions targeting heavy smokers who are already smoking throughout pregnancy. The support group studied here reduces smoking by an average of 1.18 trimesters, with an associated intent-to-treat effect of +0.433 kg on birth weight. If the reduction occurs predominantly at the first smoking margin, the true causal benefit is likely larger than what simple IV methods would suggest.

Several limitations of our analysis deserve mention. First, the exclusion restriction may be violated: the balance test shows that the support group significantly increases prenatal visits, suggesting a direct channel from the instrument to birth weight that does not operate through smoking. While the robustness check shows this has minimal quantitative impact on our estimates, it remains a conceptual concern. Second, the separability test reveals statistically significant interactions between age group and the first smoking dummy, indicating that the effect of starting to smoke may itself vary with the mother's age. If structural separability fails, the CE-TSLS estimates should be interpreted as population-weighted averages rather than single structural parameters. Third, the dataset covers births from a single Brazilian city, which limits the external validity of our findings to other populations and contexts.

Despite these limitations, the analysis demonstrates that the method of Caetano and Escanciano (2021) can recover economically meaningful and statistically distinct marginal effects in a setting where standard IV methods would be unable to separately identify them. The strong heterogeneity we document, with the first trimester of smoking being by far the most harmful, would be entirely invisible to an analyst relying on constant-effects IV, highlighting the practical value of allowing for richer treatment effect heterogeneity in program evaluation.

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